

MIAE 221: Materials Science for Engineers

Tutorial 2: Fully Solved & Annotated Master Guide

Tutorial 2 — Crystal Structures, Miller Indices, Densities & Defects (Fall 2026)



How to use this guide: Work each problem yourself first. After every question you will find a **Learn More** box pointing to the exact lecture slides and the Callister textbook expanded notes (with page numbers) where that concept is taught, so you can re-learn the theory behind any step you got wrong.

Question 1: Unit Cell, Volume & Theoretical Density

Problem Statement

Consider the unit cell of a metal shown below — atoms at the 8 corners and at the centre of each of the 6 faces.

- (a) Name the unit cell.
- (b) Derive an expression for the unit cell volume in terms of the atomic radius R .
- (c) Calculate the density of the material, given its atomic weight is **27 g/mol** and its atomic radius is **140 pm**.

Step-by-Step Solution

(a) Name of the Unit Cell

- Atoms sitting at the **8 corners** plus at the **centre of each of the 6 faces** define the **Face-Centered Cubic (FCC)** structure.
- Atoms per unit cell: corners are shared by 8 cells and face atoms by 2 cells:

$$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 1 + 3 = \mathbf{4 \text{ atoms}}$$

(b) Unit Cell Volume in Terms of R

- In FCC, neighbouring atoms **touch along the face diagonal** (corner $\rightarrow$ face-centre $\rightarrow$ corner), so the face diagonal contains **$4R$** :

$$a\sqrt{2} = 4R \implies a = \frac{4R}{\sqrt{2}} = 2\sqrt{2} R$$

- Volume of the cubic cell:

$$V_C = a^3 = \left(2\sqrt{2} R\right)^3 = 16\sqrt{2} R^3$$

(c) Theoretical Density

- Convert the radius: $R = 140 \text{ pm} = 1.40 \times 10^{-8} \text{ cm}$
- Unit cell volume:

$$V_C = 16\sqrt{2} (1.40 \times 10^{-8})^3 = 6.209 \times 10^{-23} \text{ cm}^3$$

- Theoretical density formula ($\rho = \frac{nA}{V_C N_A}$):

$$\rho = \frac{(4)(27)}{(6.209 \times 10^{-23})(6.022 \times 10^{23})} = 2.89 \text{ g/cm}^3$$

- **Sanity check:** $A \approx 27 \text{ g/mol} + \text{FCC} + \rho \approx 2.7\text{--}2.9 \text{ g/cm}^3 \rightarrow$ this metal is **aluminium (Al)**.



Learn More — Question 1

- **Teacher Lecture Notes:** *Lecture 4 — Crystal Structure 1*, **pp. 11–12** ("Face Centered Cubic Unit Cell" and "How many atoms per unit cell in the FCC structure?"). The density formula itself is worked as the "Example Problem" in *Lecture 5 — Crystal Structure 2*, **p. 8**.
- **Textbook (Expanded):** *Chapter 03 — Structure of Crystalline Solids & XRD (Expanded)*, **p. 2** (§2.2 — the FCC structure and atoms per cell) and **p. 8** (full worked example: unit cell edge length and density of an FCC metal).

Question 2: Crystallographic Directions & Planes (Miller Indices)

Problem Statement

- **(a)** Determine the indices for directions **A, B, C, D** in the cubic unit cell shown (fractional coordinates marked on the diagram).
- **(b)** Find the Miller indices for planes **A, B, C**.

Step-by-Step Solution

Method for Directions $[uvw]$

$$\text{Direction} = \text{head coordinates} - \text{tail coordinates}$$

Then multiply through to clear fractions and reduce to the **smallest integers**, enclosed in square brackets $[\]$. A negative component is written with a **bar** over the number.

(a) The Four Directions

- **Direction A:** tail $(\frac{2}{3}, 0, 1) \rightarrow$ head $(\frac{1}{2}, 1, 1)$

$$\Delta = (\frac{1}{2} - \frac{2}{3}, 1 - 0, 1 - 1) = (-\frac{1}{6}, 1, 0) \xrightarrow{\times 6} [\bar{1} \ 6 \ 0]$$

- **Direction B:** tail $(\frac{1}{3}, 1, 0) \rightarrow$ head $(1, 0, \frac{2}{3})$

$$\Delta = (\frac{2}{3}, -1, \frac{2}{3}) \xrightarrow{\times 3} [2 \ \bar{3} \ 2]$$

- **Direction C:** tail $(\frac{1}{3}, 1, 1) \rightarrow$ head $(\frac{2}{3}, 0, 0)$

$$\Delta = (\frac{1}{3}, -1, -1) \xrightarrow{\times 3} [1 \ \bar{3} \ \bar{3}]$$

- **Direction D:** tail $(\frac{1}{3}, 0, 1) \rightarrow$ head $(\frac{1}{2}, \frac{1}{2}, 0)$

$$\Delta = (\frac{1}{6}, \frac{1}{2}, -1) \xrightarrow{\times 6} [1 \ 3 \ \bar{6}]$$

Method for Planes (hkl)

1. Read the plane's **intercepts** on the x, y, z axes (in units of the lattice parameters). A plane **parallel** to an axis intercepts it at ∞ .
2. Take the **reciprocals** of the intercepts.
3. Clear fractions $\rightarrow$ smallest integers, enclosed in parentheses $()$. (If the plane passes through the origin, shift it parallel first.)

(b) The Three Planes

- **Plane A:** intercepts $(1, \infty, -1) \rightarrow$ reciprocals $(1, 0, -1) \rightarrow (1 \ 0 \ \bar{1})$
- **Plane B:** intercepts $(\frac{1}{3}, \infty, 1) \rightarrow$ reciprocals $(3, 0, 1) \rightarrow (3 \ 0 \ 1)$
- **Plane C:** intercepts $(\infty, 1, \frac{1}{2}) \rightarrow$ reciprocals $(0, 1, 2) \rightarrow (0 \ 1 \ 2)$



Learn More — Question 2

- **Teacher Lecture Notes:** *Lecture 5 — Crystal Structure 2:* directions method on **pp. 11–12** ("General Rules for Lattice Directions, Planes & Miller Indices", "Miller Indices for Directions"), worked example on **p. 13**; planes method on **pp. 18–20** ("Miller Indices for Planes").

- **Textbook (Expanded):** *Chapter 03 — Structure of Crystalline Solids & XRD (Expanded)*, p. 5 (§3 — Crystallographic Directions & Planes: Miller Indices, Callister §3.8–§3.11, including the $[uvw]$ algorithm).

Question 3: Linear & Planar Density

Problem Statement

- (a) Derive the linear density for the $[111]$ direction of FCC in terms of R .
- (b) Derive the planar density of the BCC (111) plane in terms of R .

Step-by-Step Solution

$$\text{Linear Density (LD)} = \frac{\text{number of atoms centred on the direction vector}}{\text{length of the direction vector}} \quad \text{Planar Density}$$

(a) $LD_{[111]}$ for FCC

- The $[111]$ direction is the **body diagonal**. In FCC it passes through only the **2 corner atoms** at its ends (no atom sits at the body centre), each contributing $\frac{1}{2}$:

$$\text{Atoms} = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- Length of the body diagonal = $a\sqrt{3}$. For FCC, $a = 2\sqrt{2}R$:

$$\text{Length} = 2\sqrt{2}R \times \sqrt{3} = 2\sqrt{6}R$$

$$LD_{[111]} = \frac{1}{2\sqrt{6}R} \text{ atoms per unit length}$$

(b) $PD_{(111)}$ for BCC

- The (111) plane cuts **3 corners** of the cube. Each corner atom contributes only the sector lying inside the plane's triangle: a 60° sector = $\frac{1}{6}$ of an atom:

$$\text{Atoms} = 3 \times \frac{1}{6} = \frac{1}{2} \text{ atom}$$

- The plane is an **equilateral triangle** of side $a\sqrt{2}$ (a face diagonal), so its area is:

$$A_P = \frac{\sqrt{3}}{4} (a\sqrt{2})^2 = \frac{\sqrt{3}}{2} a^2$$

- For BCC, atoms touch along the body diagonal: $a\sqrt{3} = 4R \Rightarrow a = \frac{4R}{\sqrt{3}}$, so $a^2 = \frac{16R^2}{3}$ and:

$$A_P = \frac{\sqrt{3}}{2} \times \frac{16R^2}{3} = \frac{8\sqrt{3}}{3} R^2$$

$$PD_{(111)} = \frac{1/2}{\frac{8\sqrt{3}}{3} R^2} = \frac{\sqrt{3}}{16 R^2} \text{ atoms per unit area}$$

Learn More — Question 3

- **Teacher Lecture Notes:** *Lecture 6 — Crystal Structure 3*, pp. 3–7: Linear Density defined on p. 3 with the FCC [100] worked example on p. 4; Planar Density with the FCC (110) worked example on pp. 5–6; "why do we care" on p. 7.
- **Textbook (Expanded):** *Chapter 03 — Structure of Crystalline Solids & XRD (Expanded)*, p. 6 (§3.3 — Linear & Planar Atomic Densities).

Question 4: Line Defects (Dislocations)

Problem Statement

Explain line defects and discuss its types.

Step-by-Step Solution

- **Definition:** Line defects are **one-dimensional defects** in which rows of atoms are misaligned along a line inside the crystal. They are called **dislocations**.
- **How they form:** during solidification (rapid cooling from liquid to solid), from thermal stresses, and from mechanical deformation of the crystal.

Type 1 — Edge Dislocation

- An **extra half-plane of atoms** is inserted into the crystal; the atomic planes around the end of that half-plane are bent/misaligned.
- The **Burgers vector** $\vec{b}$ is **perpendicular** to the dislocation line ($\vec{b} \perp \vec{t}$).

Type 2 — Screw Dislocation

- The crystal is **cut partway and sheared by one atomic spacing**, so the atomic planes wind around the dislocation line like a **helical ramp** (a spiral parking-garage shape — hence "screw").
- The **Burgers vector** $\vec{b}$ is **parallel** to the dislocation line ($\vec{b} \parallel \vec{t}$).

Type 3 — Mixed Dislocation

- Most real dislocations are **mixed**: the line curves through the crystal, so it has **edge character in some regions and screw character in others** — the Burgers vector lies at an angle between perpendicular and parallel to the line.

Type	Geometry	Burgers vector vs. line
Edge	Extra half-plane of atoms inserted	$\vec{b} \perp \vec{t}$ (perpendicular)
Screw	Planes spiral helically around the line	$\vec{b} \parallel \vec{t}$ (parallel)
Mixed	Curved line, both characters	At an angle

Learn More — Question 4

- **Teacher Lecture Notes:** *Lecture 7 — Defects 1*, **pp. 16–19**: "Linear Defects — Dislocations" on **p. 16**, Edge Dislocation on **p. 17**, Screw Dislocation on **p. 18**, and mixed dislocations on **p. 19**.
- **Textbook (Expanded):** *Chapter 04 — Imperfections in Solids & Defects (Expanded)*, **pp. 6–7** (§4.1 — the Burgers vector and Burgers circuit; how the dislocation character changes continuously from pure edge, $\vec{b} \perp \vec{t}$, to pure screw, $\vec{b} \parallel \vec{t}$).

Question 5: Vacancy Concentration vs. Temperature

Problem Statement

For a particular metal, there is **one vacancy every 15,000 atoms at 800°C**. At what temperature will there be **1 vacancy every 5,000 atoms**?

Step-by-Step Solution

- The equilibrium vacancy fraction follows the Arrhenius law:

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{kT}\right)$$

where Q_v is the vacancy formation energy and k is Boltzmann's constant.

- At T_1 : $T_1 = 800 + 273 = 1073$ K, $\frac{N_v}{N} = \frac{1}{15000}$:
$$-\ln(15000) = -\frac{Q_v}{kT_1} \quad (1)$$

- At T_2 : $\frac{N_v}{N} = \frac{1}{5000}$:

$$-\ln(5000) = -\frac{Q_v}{kT_2} \quad (2)$$

- **Divide (1) by (2)** — Q_v and k cancel, so the formation energy is never needed:

$$\frac{\ln(15000)}{\ln(5000)} = \frac{T_2}{T_1} = \frac{9.616}{8.517} = 1.129$$

- Solve for T_2 :

$$T_2 = 1.129 \times 1073 \text{ K} = \mathbf{1211 \text{ K}} = \mathbf{938^\circ \text{C}}$$

(In class the ratio was rounded to 1.12 before multiplying, giving $T_2 \approx 1201 \text{ K} \approx 928^\circ \text{C}$ — same method, just earlier rounding.)

- **Sanity check:** a higher temperature must give *more* vacancies, and $\frac{1}{5000} > \frac{1}{15000}$ ✓



Learn More — Question 5

- **Teacher Lecture Notes:** *Lecture 7 — Defects 1*, **pp. 3–5**: types of imperfections on **p. 3**, the vacancy as a point defect on **p. 4**, and "Equilibrium Concentration: Point Defects" (the N_v formula and its temperature dependence) on **p. 5**.
- **Textbook (Expanded):** *Chapter 04 — Imperfections in Solids & Defects (Expanded)*, **p. 3** (the $N_v = N \exp(-Q_v/kT)$ law) and **p. 9** (a fully worked equilibrium-vacancy example).

Question 6: Single Crystals vs. Polycrystals

Problem Statement

- **(a)** Explain the differences between single crystals and polycrystals.
- **(b)** Discuss how properties vary with the direction of measurement for each.

Step-by-Step Solution

(a) Structural Difference

- **Single crystal:** the whole piece of material is **one single crystal** — every unit cell in it is aligned in the **same crystallographic orientation**, with no internal boundaries.
- **Polycrystal:** the material is made of **many small crystals called grains**, each with a **different (essentially random) orientation**, joined together at **grain boundaries**.

(b) Direction-Dependence of Properties

- **Single crystal** → **ANISOTROPIC**: properties (e.g., elastic modulus, conductivity) **change with the direction** in which they are measured, because atomic packing and spacing differ along different crystallographic directions.
- **Polycrystal** → **ISOTROPIC**: the individual grains are randomly oriented, so their direction-dependent properties **average out** — the bulk material shows the **same properties in every direction**. (*Caveat: if processing aligns the grains — a "texture" — a polycrystal can become anisotropic again.*)

	Single Crystal	Polycrystal
Structure	One crystal, one orientation throughout	Many grains, random orientations
Boundaries	None	Grain boundaries between grains
Properties vs. direction	Anisotropic — vary with direction	Isotropic — same in all directions

Learn More — Question 6

- **Teacher Lecture Notes**: *Lecture 6 — Crystal Structure 3, pp. 8–9* ("Crystals As Building Blocks" — why engineering uses single crystals, e.g., turbine blades — and "Single Vs Polycrystals"); plus *Lecture 5 — Crystal Structure 2, p. 17* ("Isotropy vs. Anisotropy in Single Crystals").
- **Textbook (Expanded)**: *Chapter 04 — Imperfections in Solids & Defects (Expanded), p. 7* (§5/§4.6 — Interfacial Defects: grain boundaries separating the differently-oriented grains of a polycrystalline material).

End of Tutorial 2 — fully solved master guide. Source: MIAE 221 Tutorial 2 (Fall 2026), worked solutions as presented in class.